

Summary of Current Model Verification Study

Goal

The goal of this round was to find a concrete fermionic model that can support the perturbative SDP improvement framework in the draft. In particular, we wanted to verify three points:

1. exactness of the reduced SDP at a reference Hamiltonian H_0 ;
2. first-order response matching between the reduced SDP and the physical ground state at H_0 ;
3. a nontrivial lower bound

$$\|\delta m_S(\theta)\| \geq c_{\text{mis}}|\theta|$$

for selected moments under the perturbative Hamiltonian $H(\theta) = H_0 + \theta V$.

We tested two spinful Hubbard-type candidates, both based on an isolated Hubbard-dimer reference point.

Common Setup And Notation

- Lattice: an open one-dimensional chain with L sites, L even.
- Filling: half filling, $N_e = L$ electrons.
- In this report, $\mathcal{D}_r = (2r - 1, 2r)$, $r = 1, \dots, L/2$, denotes the two-site unit that is coupled in the reference Hamiltonian H_0 . The calligraphic symbol is used to avoid confusion with the onsite double occupancy D_i below.
- In this report, $\mathcal{B}_r = (2r, 2r + 1)$, $r = 1, \dots, L/2 - 1$, denotes the inter-unit bond on which the perturbation is applied.
- Operators:
 - $c_{i,\sigma}, c_{i,\sigma}^\dagger$: fermion annihilation/creation operators at site i with spin $\sigma \in \{\uparrow, \downarrow\}$.
 - $n_{i,\sigma} = c_{i,\sigma}^\dagger c_{i,\sigma}$, and $n_i = n_{i,\uparrow} + n_{i,\downarrow}$.
 - $D_i = n_{i,\uparrow} n_{i,\downarrow}$, the onsite double occupancy.
 - $K_{ij} = \sum_\sigma (c_{i,\sigma}^\dagger c_{j,\sigma} + c_{j,\sigma}^\dagger c_{i,\sigma})$, the bond kinetic observable.
 - $S_i^z = (n_{i,\uparrow} - n_{i,\downarrow})/2$, $S_i^+ = c_{i,\uparrow}^\dagger c_{i,\downarrow}$, and $S_i^- = c_{i,\downarrow}^\dagger c_{i,\uparrow}$.
 - *h.c.* denotes the Hermitian conjugate.

As an indexing example, when $L = 6$,

$$\mathcal{D}_1 = (1, 2), \quad \mathcal{D}_2 = (3, 4), \quad \mathcal{D}_3 = (5, 6),$$

whereas

$$\mathcal{B}_1 = (2, 3), \quad \mathcal{B}_2 = (4, 5).$$

Thus $\mathcal{D}_r$ labels the pairs already present in H_0 , while $\mathcal{B}_r$ labels the bonds connecting neighboring pairs.

The common reference Hamiltonian is the isolated-dimer spinful Hubbard model with $t_d = 1$:

$$H_0 = \sum_{r=1}^{L/2} [-K_{2r-1, 2r} + U(D_{2r-1} + D_{2r})].$$

With the same $L = 6$ indexing example, this is

$$\begin{aligned} H_0 = & [-K_{1,2} + U(D_1 + D_2)] + [-K_{3,4} + U(D_3 + D_4)] \\ & + [-K_{5,6} + U(D_5 + D_6)]. \end{aligned}$$

For response matching, fields are coupled to the selected moments $M_1, \dots, M_s$ by

$$H(\theta, h) = H_0 + \theta V - \sum_{a=1}^s h_a M_a.$$

Reported Quantities

The quantities in this subsection are diagnostics used in this numerical report, not standard names from the Hubbard-model literature.

- Reference energy error: $|E_{\text{SDP}}(H_0) - E_{\text{phys}}(H_0)|$.
- Selected-moment error: $\|m_S^{\text{SDP}}(0) - m_S^{\text{phys}}(0)\|$.
- Response-matching error: $\|D_h m_S^{\text{SDP}}(0) - D_h m_S^{\text{phys}}(0)\|$, the norm of the difference between the SDP and physical first-order response matrices.
- Mismatch ratio: $\|\delta m_S(\theta)\|/|\theta|$, where $\delta m_S(\theta) = m_S^{\text{SDP}}(\theta) - m_S^{\text{phys}}(\theta)$. This ratio is used to estimate c_{mis} .
- Solver: all reported SDP values below were computed with MOSEK.

As a concrete example, if the selected moments are $M_r = K_{2r-1, 2r}$ and $L = 6$, then

$$m_S = (\langle K_{1,2} \rangle, \langle K_{3,4} \rangle, \langle K_{5,6} \rangle).$$

A response-matching error measures how close the SDP and exact physical first-order changes of this vector are when small fields h_a are applied. A mismatch ratio near 0.21 means that the SDP and physical selected-moment vectors differ by about $0.21|\theta|$ in norm over the tested range.

Terminology And References

- The terms “Hubbard model”, “spinful Hubbard chain”, “hopping”, “onsite interaction”, “half filling”, and “double occupancy” follow the standard Hubbard-model terminology; see Hubbard’s original paper and standard treatments of the one-dimensional Hubbard model [1, 2].
- The words “dimerized” and “alternating” refer to chains with alternating bond strengths; this terminology is standard in the dimerized or alternating Hubbard-chain literature [3, 4].
- The term “extended Hubbard model” refers here to adding an intersite density-density interaction to the onsite Hubbard interaction; this is the standard usage in the extended-Hubbard literature [5].
- “Response matrix” is used in the usual static linear-response sense, following the Kubo linear-response terminology [6].
- “Semidefinite programming” follows the standard convex-optimization terminology of Vandenberghe and Boyd [7].

Model 1: Standard Dimerized Spinful Hubbard Chain

Model. The first model is the standard alternating-hopping spinful Hubbard chain. It uses the common reference Hamiltonian H_0 above and the inter-dimer hopping perturbation

$$V_{\text{hop}} = - \sum_{r=1}^{L/2-1} K_{2r, 2r+1}.$$

The reported scan used $L = 4$, $U = 2$, $t_d = 1$, and $\theta = \pm 0.005, \pm 0.01$.

The selected moment families tested were:

- Bond kinetic moments within $\mathcal{D}_r$: $M_r = K_{2r-1,2r}$.
- Double occupancy moments within $\mathcal{D}_r$: $M_r = D_{2r-1} + D_{2r}$, and also the site-resolved versions D_{2r-1} , D_{2r} .
- Spin correlation moments within $\mathcal{D}_r$: $S_{2r-1}^z S_{2r}^z$, and $\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r} = S_{2r-1}^z S_{2r}^z + \frac{1}{2}(S_{2r-1}^+ S_{2r}^- + S_{2r-1}^- S_{2r}^+)$.
- Onsite-pair hopping moments between the two sites of $\mathcal{D}_r$: $P_r = c_{2r-1,\uparrow}^\dagger c_{2r-1,\downarrow}^\dagger c_{2r,\downarrow} c_{2r,\uparrow} + h.c.$.
- Singlet/triplet projectors on $\mathcal{D}_r$: $\Pi_{s,r} = |s_r\rangle\langle s_r|$ and $\Pi_{t0,r} = |t_r^0\rangle\langle t_r^0|$, where $|s_r\rangle = (|\uparrow, \downarrow\rangle - |\downarrow, \uparrow\rangle)/\sqrt{2}$ and $|t_r^0\rangle = (|\uparrow, \downarrow\rangle + |\downarrow, \uparrow\rangle)/\sqrt{2}$ on $\mathcal{D}_r$.
- Bond kinetic moments on $\mathcal{B}_r$: $M_r = K_{2r,2r+1}$.

As an indexing example, when $L = 6$, the hopping perturbation is

$$V_{\text{hop}} = -(K_{2,3} + K_{4,5}),$$

which acts on $\mathcal{B}_1$ and $\mathcal{B}_2$. In the same $L = 6$ system, the bond kinetic moments within the reference units are

$$K_{1,2}, \quad K_{3,4}, \quad K_{5,6},$$

whereas the bond kinetic moments on the perturbation bonds are

$$K_{2,3}, \quad K_{4,5}.$$

Verification results.

1. **Reference exactness.** The reference SDP was solved at H_0 . The following rows give representative moment choices from the scan.

Selected moments	Energy error	Moment error	Status
$K_{2r-1,2r}$	1.47×10^{-9}	9.37×10^{-8}	OPTIMAL
$D_{2r-1} + D_{2r}$	1.47×10^{-9}	4.74×10^{-8}	OPTIMAL
$K_{2r,2r+1}$	1.47×10^{-9}	4.65×10^{-16}	OPTIMAL

2. **First-order response matching.** Moments supported within $\mathcal{D}_r$ had small response-matching errors. The bond kinetic moment on $\mathcal{B}_r$ had a large response-matching error.

Selected moments	Support	Response error
$K_{2r-1,2r}$	within $\mathcal{D}_r$	4.86×10^{-7}
$D_{2r-1} + D_{2r}$	within $\mathcal{D}_r$	1.56×10^{-7}
$\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r}$	within $\mathcal{D}_r$	1.44×10^{-7}
$\Pi_{s,r}$	within $\mathcal{D}_r$	1.10×10^{-7}
$K_{2r,2r+1}$	on $\mathcal{B}_r$	7.05

3. **Mismatch lower bound.** For moments supported within $\mathcal{D}_r$, the ratio $\|\delta m_S(\theta)\|/|\theta|$ approximately doubled when $|\theta|$ doubled from 0.005 to 0.01. This indicates quadratic, not linear,

scaling in the tested range. The bond kinetic moment on $\mathcal{B}_r$ had a stable linear ratio, but it failed the response-matching check above.

Selected moments	Support	Ratio at 0.005	Ratio at 0.01	Observation
$K_{2r-1,2r}$	within $\mathcal{D}_r$	5.96×10^{-3}	1.17×10^{-2}	quadratic
$D_{2r-1} + D_{2r}$	within $\mathcal{D}_r$	7.04×10^{-3}	1.40×10^{-2}	quadratic
$\mathbf{S}_{2r-1} \cdot \mathbf{S}_{2r}$	within $\mathcal{D}_r$	6.36×10^{-3}	1.26×10^{-2}	quadratic
P_r	within $\mathcal{D}_r$	5.42×10^{-3}	1.09×10^{-2}	quadratic
$K_{2r,2r+1}$	on $\mathcal{B}_r$	4.59	4.59	linear ratio, response failed

For example, the first row says that for the selected vector built from $K_{2r-1,2r}$, the mismatch ratio increased from 5.96×10^{-3} to 1.17×10^{-2} when $|\theta|$ doubled.

Model 2: Density-Perturbed Extended Spinful Hubbard Dimer Chain

Model. The second model uses the same isolated Hubbard-dimer reference H_0 , but replaces the hopping perturbation by an inter-dimer density-density perturbation:

$$V_{\text{dens}} = \sum_{r=1}^{L/2-1} n_{2r} n_{2r+1}.$$

The selected moments are the bond kinetic observables within $\mathcal{D}_r$:

$$M_r = K_{2r-1,2r}, \quad r = 1, \dots, L/2.$$

As an indexing example, when $L = 6$,

$$V_{\text{dens}} = n_2 n_3 + n_4 n_5,$$

while the selected moments are

$$K_{1,2}, \quad K_{3,4}, \quad K_{5,6}.$$

Thus this model perturbs the links $\mathcal{B}_1, \mathcal{B}_2$ but measures bond kinetic observables inside $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$.

The reported checks used $U = 4$, $t_d = 1$, and $L = 4, 6$.

Verification results.

1. **Reference exactness.** At H_0 , both the SDP energy and selected bond-kinetic moments matched the exact ground-state values to high precision.

System	Energy error	Moment error	Status
$L = 4, U = 4$	5.74×10^{-12}	1.01×10^{-7}	OPTIMAL
$L = 6, U = 4$	8.18×10^{-13}	1.03×10^{-9}	OPTIMAL

2. **First-order response matching.** The SDP and physical response matrices were compared by finite differences with field step $h = 0.005$.

System	Selected moments	Response error
$L = 4, U = 4$	$K_{2r-1, 2r}$	3.04×10^{-5}
$L = 6, U = 4$	$K_{2r-1, 2r}$	3.46×10^{-6}

3. **Mismatch lower bound.** The ratio $\|\delta m_S(\theta)\|/|\theta|$ stayed nearly constant over the tested θ -range, giving a numerical lower-bound constant.

System	Tested $ \theta $	Ratio range	Conservative c_{mis}
$L = 4, U = 4$	0.002, 0.005, 0.01, 0.02	0.124–0.125	0.12
$L = 6, U = 4$	0.005, 0.01	0.215–0.216	0.21

For example, the $L = 6, U = 4$ row says that the observed mismatch ratio was about 0.215 at both tested values of $|\theta|$, so 0.21 was used as a conservative numerical lower-bound constant on this tested range.

Observed Status

For the standard hopping perturbation, the reference exactness error was small, the response check for moments supported within $\mathcal{D}_r$ passed numerically, and the corresponding mismatch was quadratic in θ in the tested cases.

For the density-density perturbation, the tested $L = 4$ and $L = 6$ cases both gave small reference errors, small response-matching errors, and a linear mismatch ratio bounded away from zero on the tested θ -range.

References

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